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Sub ject : DSA

Unit 3, S2, SLO1 - Stack Implementation Using Array - Advantages, Disadvantages and Applications

1. Stack implementation using arrays is ____ to implement.

Answer: easy / simple

2. Accessing elements in an array is ____ due to direct indexing.

Answer: fast (constant time $O(1)$)

3. Stack operations like push and pop take ____ time in array implementation.

Answer: $O(1)$ constant

4. Arrays provide ____ memory allocation at compile time.

Answer: static

5. Array-based stacks have ____ memory access, improving performance.

Answer: contiguous / cache-friendly

6. There is no overhead of memory allocation and ____ in arrays.

Answer: deallocation (pointer overhead)

7. Array-based stack implementation avoids the extra memory used for ____ in linked lists.

Answer: pointers (next address fields)

8. It is easier to ____ a stack implemented using arrays in programming.

Answer: debug and visualize

9. Array implementation of stack is ____ in terms of memory for small, fixed-size stacks.

Answer: efficient

10. Stack overflow in array implementation is ____ to detect.

Answer: straightforward / easy (check $TOP == MAX-1$)

11. Stack implemented using arrays has a fixed ____ size.

Answer: maximum capacity

12. If the stack size is too small, it may lead to ____.

Answer: stack overflow

13. If the stack size is too large, it may lead to ____ of memory.

Answer: wastage (unused allocated space)

14. Arrays use ____ memory allocation, which is less flexible.

Answer: static

15. It is difficult to ____ the size of a stack implemented using arrays at runtime.

Answer: resize / dynamically change

16. Stack operations using arrays are not memory ____ for large data.

Answer: flexible / dynamic

17. Inserting or removing elements from the stack may require checking for ____ or underflow.

Answer: overflow

18. Reallocation of array memory is ____ and not always supported in some languages.

Answer: costly ($O(n)$ copying)

19. Array implementation is not suitable when the maximum stack size is ____.

Answer: unknown or unpredictable

20. Stack implemented using arrays may not efficiently use ____ in dynamic scenarios.

Answer: memory resources

21. Stacks are used in ____ evaluation and conversion.

Answer: expression (infix, postfix, prefix)